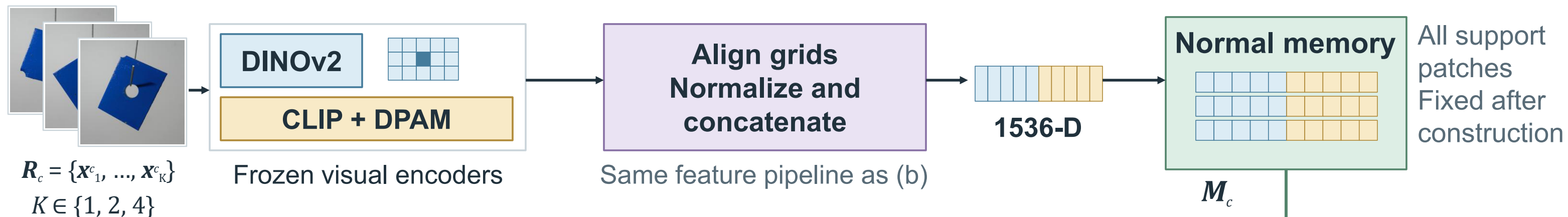


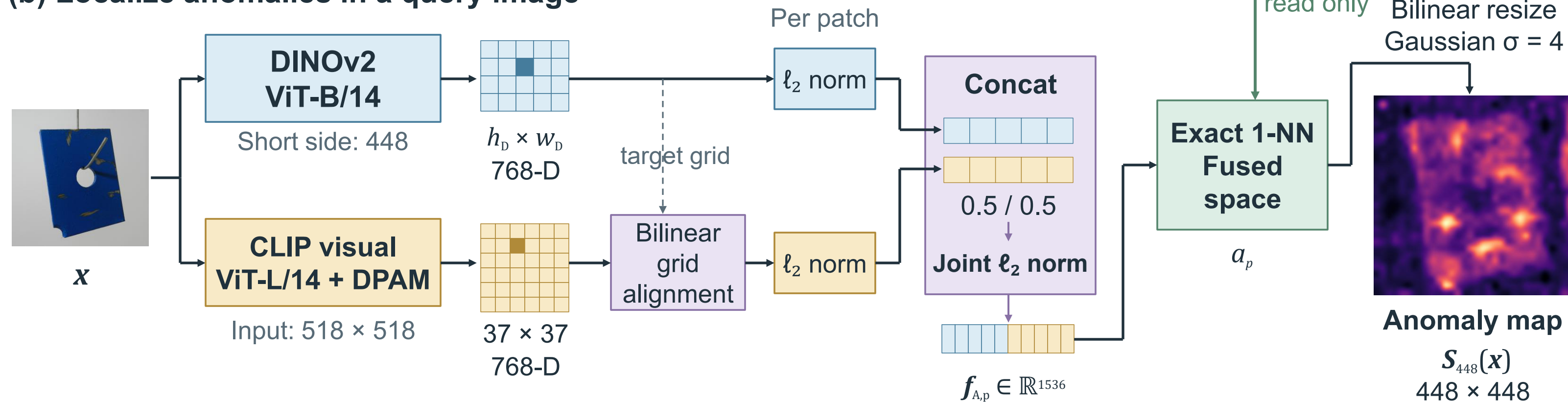
DCFnet Dual-Encoder Calibrated Fusion Network

Normal-only few-shot industrial anomaly localization

(a) Build the normal reference memory



(b) Localize anomalies in a query image



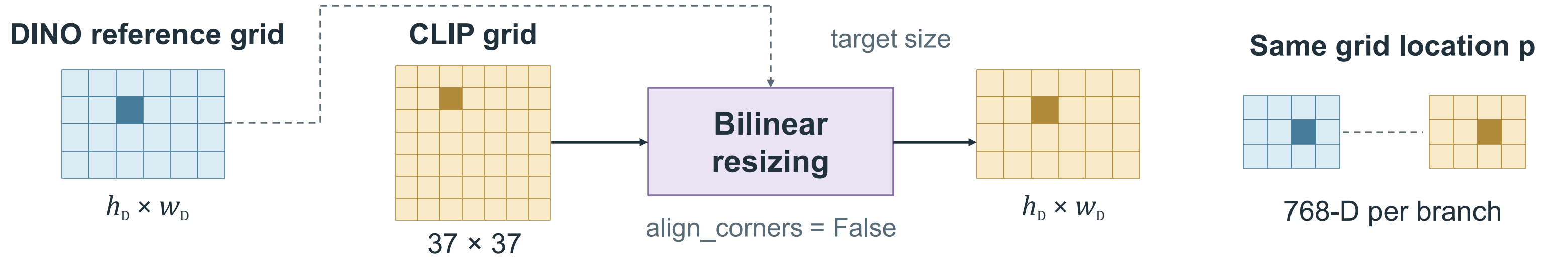
$$a_p = \frac{1}{2} \min_{m \in M_c} \|f_{A,p}(x) - m\|_2^2$$

$$s_{\text{img}}(x) = \max_u S_{448,u}(x)$$

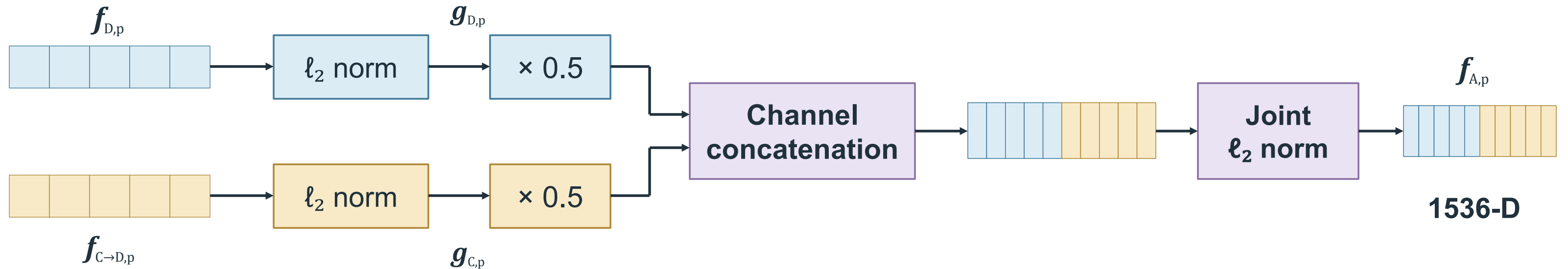
Fig. 2 Spatial calibration and fixed feature fusion

Common spatial sampling, normalized branch magnitudes, and a joint descriptor

(a) Align the CLIP grid to the DINO lattice



(b) Normalize, scale and concatenate each patch pair

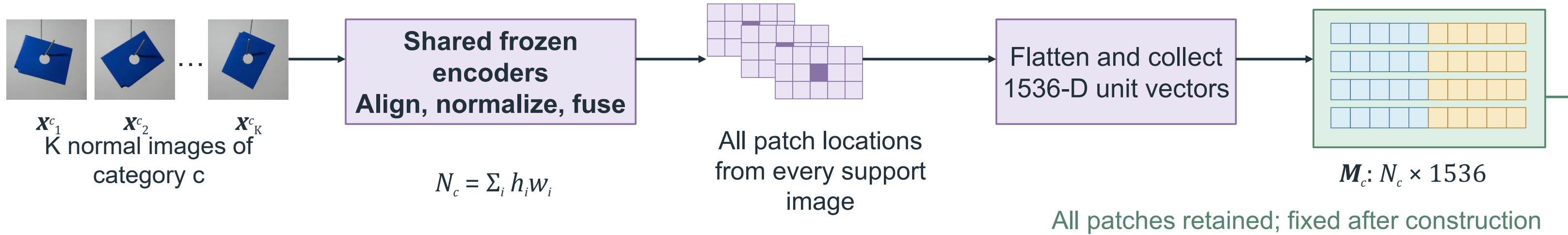


$$f_{A,p} = \text{Norm}([0.5 \mathbf{g}_{D,p}; 0.5 \mathbf{g}_{C,p}]) = [\mathbf{g}_{D,p}; \mathbf{g}_{C,p}] / \sqrt{2}$$

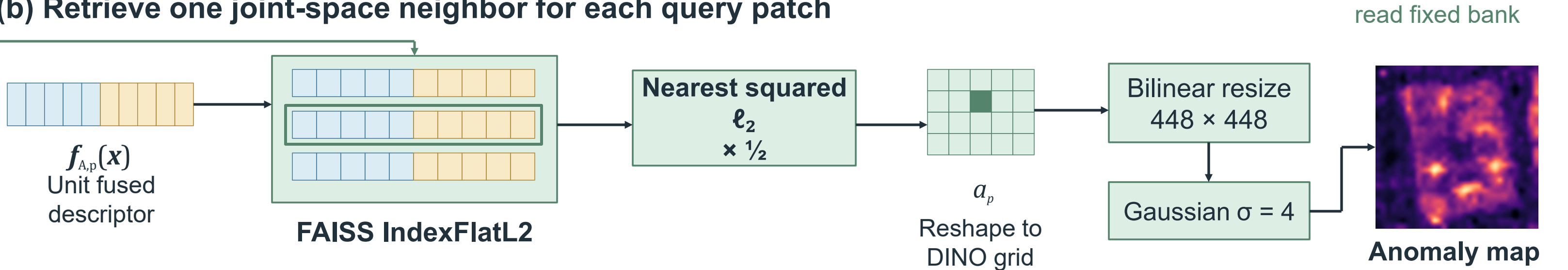
Fig. 3 Normal memory construction and exact anomaly scoring

Category-specific support patches define one fixed search space for every query patch

(a) Build the category-specific memory once



(b) Retrieve one joint-space neighbor for each query patch



$$d_{pq} = \frac{1}{2} \|\mathbf{g}_{D,p} - \mathbf{g}_{D,q}\|_2^2 + \frac{1}{2} \|\mathbf{g}_{C,p} - \mathbf{g}_{C,q}\|_2^2$$

Both branches evaluate the same support candidate q

$$a_p = \frac{1}{2} \min_q d_{pq}$$

$$s_{\text{img}}(\mathbf{x}) = \max_u \mathcal{S}_{448,u}(\mathbf{x})$$

Fig. S1 Frozen dual-encoder feature extraction

The same visual pipeline encodes every normal support image and each query image

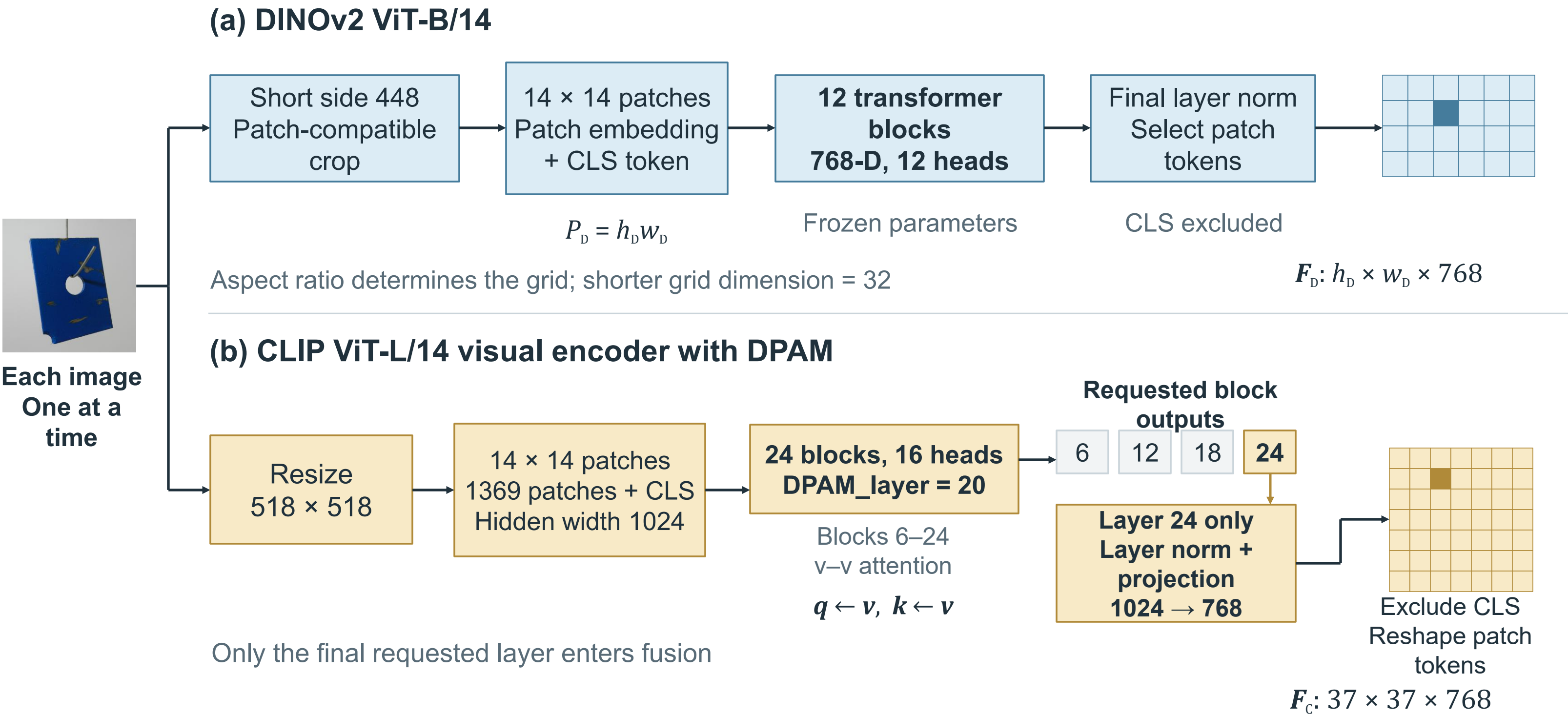
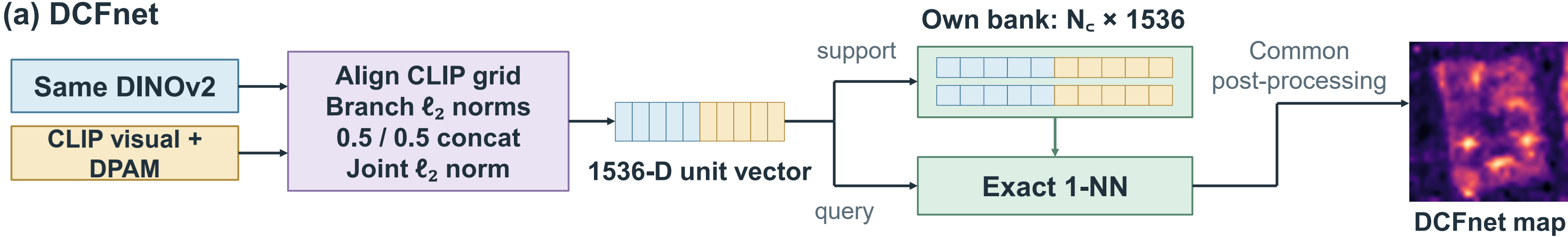


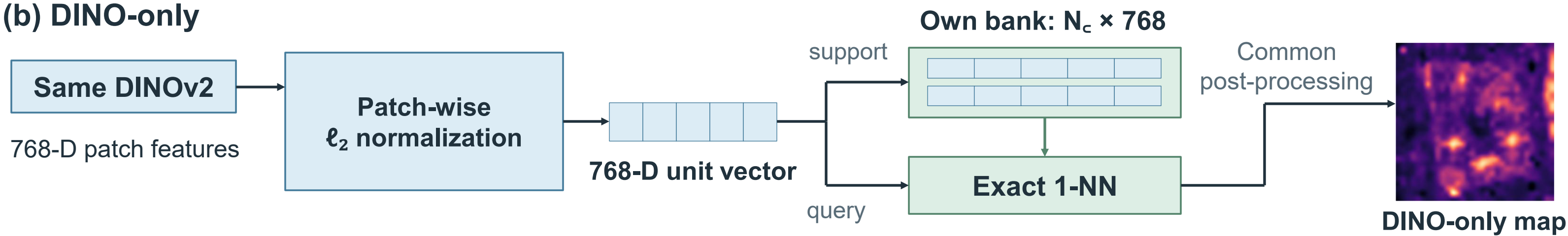
Fig. S2 Matched control for the added visual representation

Paired inputs and a common scorer isolate the effect of adding calibrated CLIP features

(a) DCFnet



(b) DINO-only



(c) Fixed conditions in every paired run

Same input manifest

K = 1, 2, 4
Seeds = 0, 1, 2
Same support and query IDs

Same distance

All normal support patches
FAISS IndexFlatL2
Nearest squared ℓ_2 / 2

Same output processing

Bilinear resize to 448×448
Gaussian $\sigma = 4$
Spatial max for image score

Same evaluation

Same category and test split
Same metric implementation
Paired comparison per K / seed

Descriptor storage: DCFnet = 2 $\times$ DINO-only for the same number of stored patches

Maps: MPDD, seed 0, K = 1